(ST 651) Linear Model Theory I
Final Project

1 Introduction

This report investigates the ideas presented in the paper *"Testing Against a High Dimensional Alternative"* by Goeman et al. The framework considered is primarily the linear regression model:

$$\mathbf{Y}_{(n\times 1)} = \mathbf{X}_{(n\times(p+1))}\boldsymbol{\beta}_{((p+1)\times 1)} + \boldsymbol{\epsilon}_{(n\times 1)}, \qquad \boldsymbol{\epsilon} \sim N\left(0, \sigma^2 \boldsymbol{I}\right)$$

The classical approach focuses on the F-test to examine

$$H_0 : \boldsymbol{\beta}_{(-1)} = 0 \quad \text{versus} \quad H_A : \boldsymbol{\beta}_{(-1)} \neq 0,$$

i.e., whether any regression coefficient (excluding intercept) is nonzero. The resulting F-statistic is

$$F = \frac{(\mathrm{RSS}_{\mathrm{Red}} - \mathrm{RSS}_{\mathrm{Full}})/p}{\mathrm{RSS}_{\mathrm{Full}}/(n-p-1)} \sim F_{p,\, n-p-1}$$

However, in high-dimensional settings ($p > n$), the classical F-test fails. Specifically, when $p > n$, $\mathbf{X}^\mathsf{T}\mathbf{X}$ is singular and the F-statistic becomes undefined. Even when $p \approx n$, residual degrees of freedom $n - p - 1$ are small, making the variance estimator $\mathrm{RSS}_{\mathrm{Full}}/(n-p-1)$ unstable and leading to low power. Some basic simulation results are presented in the Appendix illustrating such. Furthermore, as the authors note, in high dimensions the set of possible alternatives becomes extremely broad. Some alternatives may yield the same distribution of $\mathbf{y}$ as the null, making them statistically indistinguishable. Consequently, no test can uniformly maintain high power across all alternatives, and any minimax strategy is doomed to fail.

2 Method

To address these challenges, the authors propose an alternative approach using a score test applied to a hyperparameter in an empirical Bayesian framework. The idea is to concentrate the power of the test on alternatives deemed most relevant. This is achieved by assigning a prior distribution to $\boldsymbol{\beta}$ that places probability mass on plausible or interesting alternatives. The specific distribution on $\boldsymbol{\beta}$ is left unspecified, but the mean and covariance are parameterized as

$$E(\boldsymbol{\beta}) = \mathbf{0} \qquad E(\boldsymbol{\beta}\boldsymbol{\beta}^T) = \tau^2\boldsymbol{\Sigma}$$

for some "well-chosen" (i.e., identity) positive (semi)-definite $p \times p$ matrix $\boldsymbol{\Sigma}$.

Using the chosen distribution of $\boldsymbol{\beta}$, the testing problem can be reformulated in terms of the marginal distribution of $\mathbf{y}$. The marginal density is

$$\bar{f}(\tau^2 \mid \mathbf{y}) = \mathrm{E}_{\boldsymbol{\beta}|\tau^2}\{f(\boldsymbol{\beta} \mid \mathbf{y})\},$$

where $f(\boldsymbol{\beta} \mid \mathbf{y})$ is the likelihood of $\boldsymbol{\beta}$ given $\mathbf{y}$, and $\mathrm{E}_{\boldsymbol{\beta}|\tau^2}(\cdot)$ denotes expectation with respect to the distribution of $\boldsymbol{\beta}$ conditional on τ^2. This can be interpreted as the likelihood for τ^2 under a marginal model for $\mathbf{y}$. In this framework, testing

$$H_0 : \tau^2 = 0 \quad \text{versus} \quad H_A : \tau^2 = \tau_1^2$$

is equivalent to testing the original hypothesis about $\boldsymbol{\beta}$. A direct application of empirical Bayes Neyman-Pearson is complicated by the p-dimensional integral involved in the marginal likelihood and by the need to fully specify the distribution of $\boldsymbol{\beta}$ (up to τ^2). As the authors note:

> This means specifying whether the relevant alternatives correspond to a $\boldsymbol{\beta}$ vector containing a few large components or many small ones. Making this distinction is generally very challenging in high-dimensional settings. For instance, in a high-dimensional regression model, it is often unclear whether the true signal consists of a small number of large regression coefficients or a large number of small ones. An incorrect specification of the distribution of $\boldsymbol{\beta}$ may therefore lead to low statistical power.

To circumvent these difficulties, the authors recommend testing

$$\bar{H}_0 : \tau^2 = 0 \quad \text{versus} \quad \bar{H}_A : \tau^2 > 0$$

in the marginal model using the score test statistic

$$S = \frac{d}{d\tau^2} \log\{\bar{f}(0 \mid \mathbf{y})\}$$

Specifically in the linear model setting, they provide lemmas showing that the score statistic can be computed using simple matrix operations, avoiding the high-dimensional integral,

$$S = \frac{\mathbf{y}'\mathbf{X}\mathbf{X}'\mathbf{y}}{\mathbf{y}'\mathbf{y}}$$

and that it does not require specifying the exact distributional shape of $\boldsymbol{\beta}$. There is an alternative representation of the test statistic obtained from the spectral decomposition

$$\mathbf{X}\mathbf{X}' = \sum_{i=1}^{n} \lambda_i \mathbf{P}_i,$$

which yields

$$S = \sum_{i=1}^{n} \lambda_i \frac{\mathbf{y}'\mathbf{P}_i\mathbf{y}}{\mathbf{y}'\mathbf{y}}$$

This expression represents the test statistic as a weighted sum of quadratic forms $\mathbf{y}'\mathbf{P}_i\mathbf{y}$, each corresponding to a test of whether the ith principal component of $\mathbf{X}$ is associated with $\mathbf{y}$. The weights (eigenvalues) λ_i are proportional to the variances of the principal components.

Viewed in this way, the procedure resembles a weighted principal components regression, in which evidence for association is aggregated across all principal components. The difference between principal component regression and the proposed score test is that the latter incorporates the variance in each direction as weights, and avoids the task of selecting how many components to retain.

3 Discussion

With the general outline and motivation of the authors' proposed test established above, we can now turn to several reflections on both the paper and the broader problem of inference in high-dimensional settings. While the increasing availability of high-dimensional data explains the growing interest in such problems, it is less clear that classical hypothesis testing remains the most meaningful inferential objective in these contexts. In particular, when the dimension of the parameter space greatly exceeds the available sample size, I'm skeptical of the practical use in testing global null hypotheses as they may provide limited insight into the underlying structure of the data.

A practical difficulty arises immediately when attempting to implement the proposed score test. Although the previous section describes the construction of the test statistic, its null distribution is not available in closed form in general. Without knowledge of the null distribution, the test cannot be directly applied. The authors acknowledge this limitation, noting that aside from having mean zero under the null hypothesis, the statistic S is not standardized and should not be expected to follow a familiar textbook distribution. Standard asymptotic arguments are difficult to apply because the marginal likelihood $\bar{f}$, from which the score statistic is derived, is not generally a product of independent contributions across observations. In principle, one could approximate the null distribution by simulation by generating responses $\mathbf{Y}$ under the null model with random errors while holding $\mathbf{X}$ fixed, thereby approximating the sampling distribution of the statistic in order to obtain critical values.

Within the simplified linear model framework, the presented lemmas establish optimality properties of the proposed test, particularly that the score test achieves the optimal slope of the power function against local alternatives close to the null hypothesis. However, these optimality results depend heavily on the assumed prior structure placed

on $\boldsymbol{\beta}$. In particular, the choice of prior distribution effectively determines where the test will have power. The test is constructed by placing probability mass on alternatives that are deemed plausible, and consequently the resulting procedure is most powerful precisely in those regions. In this sense, the power analysis becomes somewhat circular: the procedure performs well against the alternatives that were encoded in the prior specification. In essence, we assume we know where the interesting alternative are and then test for their existence.

Roughly speaking, the test performs well when the covariance structure assumed for $\boldsymbol{\beta}$ aligns with the covariance structure of the predictors. When directions of large variation in the predictors correspond to directions in which the components of $\boldsymbol{\beta}$ are expected to be large, the test can achieve good power. However, this alignment need not occur in practice. For example, if predictors with relatively small variance happen to have large associated regression coefficients, the proposed test will have little power to detect the signal.

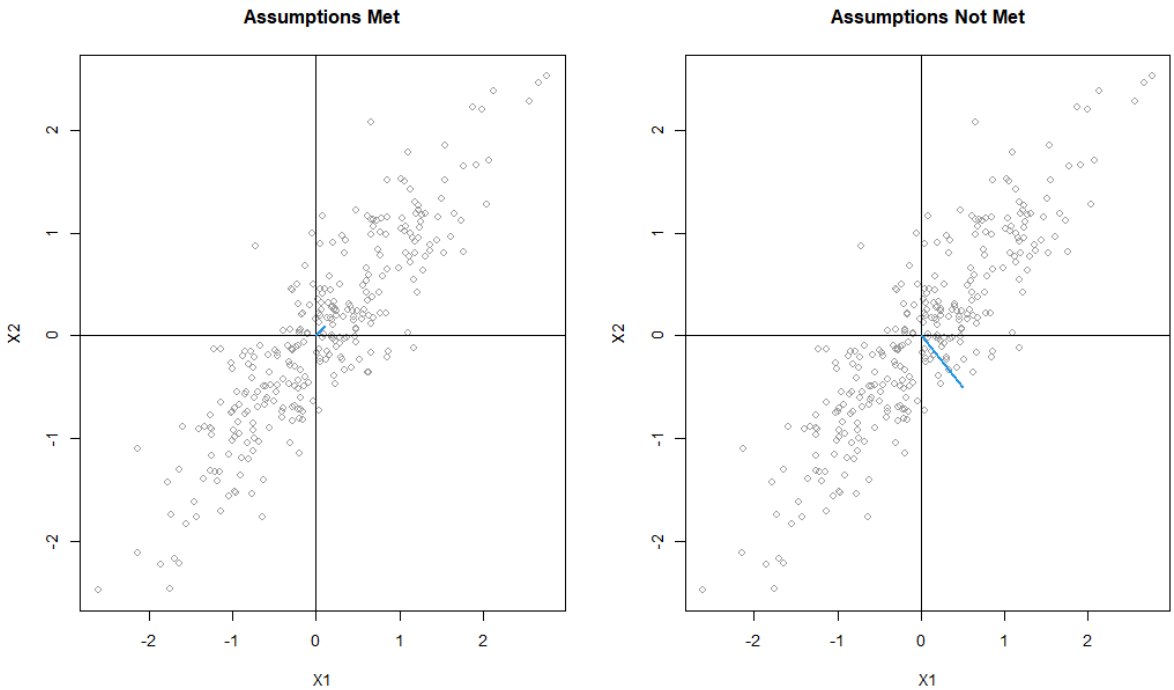

Figure 1: Illustration of alignment between predictor variation and the direction of $\boldsymbol{\beta}$. The left panel shows a case where the coefficient vector aligns with the principal direction of variation in $\mathbf{X}$, while the right panel shows a case where this alignment does not occur.

To illustrate this point, we simulate data in a simple setting with $p = 2$ predictors and a fixed design matrix $\mathbf{X}$ exhibiting positive correlation (simulation details are provided in the Appendix). This setting allows the geometry of both the predictor space and the coefficient vector $\boldsymbol{\beta}$ to be easily visualized. Figure 1 contrasts a case in which the direction of $\boldsymbol{\beta}$ aligns with the dominant variation in the predictors with a case in which it does not.

$\boldsymbol{\beta}$	F-Test Power	Score Test Power
$\begin{pmatrix}0.10 & 0.10\end{pmatrix}$	0.73126	0.83634
$\begin{pmatrix}0.50 & -0.50\end{pmatrix}$	0.92465	0.09600

Table 1: Empirical power comparison between the classical F-test and the proposed score test under two coefficient configurations (100,000 simulations were performed to estimate the power of these tests)

In the aligned case, we reduce the magnitude of $\boldsymbol{\beta}$ so that the power of both tests does not trivially approach one.

Under this configuration, the score test achieves slightly higher power than the classical F-test. In contrast, when the coefficient vector is oriented in a direction that does not align with the dominant variation in the predictors, the performance of the score test deteriorates substantially. In this scenario, the F-test retains high power while the score test fails to detect the signal. This example highlights the sensitivity of the proposed procedure to the structural assumptions encoded in the prior specification.

Another limitation concerns invariance. In classical linear models with $p < n$, the F-test possesses desirable invariance properties under reparameterizations and linear transformations of the predictor space. In contrast, the locally most powerful test derived in this framework is not invariant under such changes of parameterization. The authors explicitly note that the F-test optimizes power over ellipsoidal sets of alternatives defined through $\boldsymbol{\beta}'\mathbf{X}'\mathbf{X}\boldsymbol{\beta}$, whereas the proposed score test optimizes power over regions defined by $\boldsymbol{\beta}'\boldsymbol{\beta}$. As a consequence, the geometry of the predictor space—captured by $\mathbf{X}'\mathbf{X}$—is not incorporated into the construction of the test thus simple rescalings or linear transformations of the predictors can substantially alter the behavior of the test.

This lack of invariance is particularly relevant in high-dimensional settings. When $p > n$, certain directions in parameter space are weakly or completely unidentifiable from the data. In such cases, no nontrivial invariant test exists, since the observed data contain no information about some directions of $\boldsymbol{\beta}$.

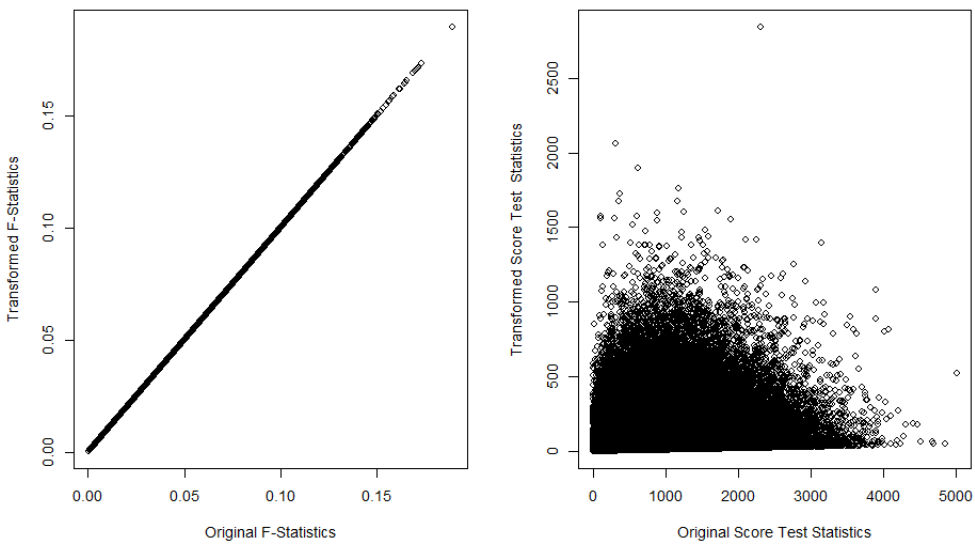

Figure 2: Comparison of the F-test and score test statistics before and after a linear transformation of the predictor space.

To illustrate the practical consequences of this lack of invariance, we consider a simple simulation study (detailed outline in the Appendix). Starting from the linear model outlined before, we apply a linear transformation to the predictors so that

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon} = \underbrace{\mathbf{X}\mathbf{A}}_{\mathbf{Z}}\underbrace{\mathbf{A}^{-1}\boldsymbol{\beta}}_{\boldsymbol{\gamma}} + \boldsymbol{\epsilon} = \mathbf{Z}\boldsymbol{\gamma} + \boldsymbol{\epsilon}, \qquad \text{where} \qquad \mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/10 & 0 \\ 0 & 0 & 10 \end{pmatrix}$$

Thus the intercept column is unchanged, while the first predictor is rescaled by a factor of $1/10$ and the second predictor by a factor of 10. The corresponding regression coefficients are adjusted through $\boldsymbol{\gamma} = \mathbf{A}^{-1}\boldsymbol{\beta}$ so that the underlying model for $\mathbf{Y}$ remains unchanged.

Figure 2 displays scatterplots comparing the resulting test statistics computed before and after the transformation. Because the F-test is invariant under linear transformations of the predictors, the transformed and untransformed

statistics exhibit a one-to-one (identity) relationship. In contrast, the score statistics show no such relationship, instead forming a diffuse cloud of points.

	F-Test Power	Score Test Power
$\boldsymbol{\beta} = \begin{pmatrix} 0.02 & 0.02 \end{pmatrix}$	0.81476	0.91948
$\boldsymbol{\gamma} = \begin{pmatrix} 0.20 & 0.002 \end{pmatrix}$	0.81476	0.08326

Table 2: Empirical power comparison between the classical F-test and the proposed score test under two equivalent parameterizations. Power was estimated using 100,000 simulations.

The power results further illustrate this phenomenon. As expected, the F-test yields identical power under the two parameterizations, reflecting its invariance to linear transformations of the predictors. In contrast, the power of the score test changes dramatically, decreasing from approximately 0.92 to 0.08 under the transformed parameterization. Although the two models generate identical distributions for $\mathbf{Y}$, the score test behaves very differently depending on how the predictors are scaled. This example highlights the practical implications of the lack of invariance: the performance of the proposed procedure can depend strongly on arbitrary choices of parameterization or predictor scaling.

Lastly, I found the selection of assumptions permitted within the framework somewhat inconsistent. As quoted in the discussion of the method's formulation, the proposed score test avoids specifying the full distribution of $\boldsymbol{\beta}$, with the authors acknowledging that it is risky to assume the unknown distribution of the parameters of interest can be accurately specified. However, the analysis is still built upon several other structural assumptions about $\boldsymbol{\beta}$. In particular, the components of $\boldsymbol{\beta}$ are treated as exchangeable and a covariance structure proportional to the identity matrix, $\boldsymbol{\Sigma} = \mathbf{I}$, is adopted, though $\mathbf{I}$ is just one special case of an exchangeable covariance matrix.

As discussed previously, the power of the proposed test depends directly on how this prior structure is specified, as well as on an implicit alignment between the variance structure of the predictors and the magnitude and direction of the components of $\boldsymbol{\beta}$. Consequently, while the method avoids one type of distributional specification, it still relies on other assumptions that encode strong beliefs about the structure of the regression coefficients. In practice, it is not clear that these assumptions are substantially easier to justify than the distributional assumptions the authors aim to avoid.

4 Conclusion

There remain several additional aspects of this paper that warrant further investigation, alas the page limit of report is nearing. As discussed earlier, the proposed statistic can also be interpreted as a form of weighted principal components regression. One natural extension of the present analysis would therefore be to compare the performance of the proposed score test directly with standard principal components regression, in order to assess whether the weighting scheme used by the authors provides a meaningful improvement. It would also be informative to compare the procedure with other competing approaches mentioned in the paper, such as the classical F-test and the maximal T-test.

Another potential avenue of investigation concerns the treatment of the design matrix $\mathbf{X}$. Throughout the paper and in the accompanying simulations, the authors condition on $\mathbf{X}$ and treat it as fixed rather than as a random quantity. However, in many practical applications the predictors themselves are naturally viewed as random variables. The described score test is very sensitive to a change in the design matrix, so investigating behaviors such with a random $\mathbf{X}$ might reveal even more noteworthy features (bugs?) of this test.

Overall, while the paper provides an interesting perspective on hypothesis testing in high-dimensional linear models, the practical performance of the proposed test depends strongly on structural assumptions about the coefficient vector and the scaling of the predictors. As the simulations illustrate, departures from these assumptions can substantially affect the behavior of the procedure, suggesting that careful consideration is required before applying the method in practice.

5 Appendix

5.1 F-Test Power

The following simulation study examines the empirical power of the overall F-test in a linear regression model as the number of predictors and the magnitude of the regression coefficients vary. Data were generated according to the linear model

$$\mathbf{Y}_{(n\times 1)} = \mathbf{X}_{(n\times(p+1))}\boldsymbol{\beta}_{((p+1)\times 1)} + \boldsymbol{\epsilon}_{(n\times 1)}, \qquad \boldsymbol{\epsilon} \sim N\big(0, \sigma^2\mathbf{I}_n\big)\,,$$

where

$$n = 100, \qquad \sigma^2 = 1, \qquad \alpha = 0.05$$

The design matrix $\mathbf{X}$ consisted of an intercept column and p predictors generated independently from a standard normal distribution,

$$X_{ij} \overset{\text{iid}}{\sim} N(0,1)$$

The coefficient vector was defined as

$$\boldsymbol{\beta} = (1, \beta, \beta, \ldots, \beta)^{\mathsf{T}},$$

where the intercept was fixed at 1 and the remaining p coefficients shared a common magnitude β. For each combination of p and β, 10,000 datasets were simulated. For each simulated dataset, the overall F-statistic testing $H_0 : \beta_1 = \beta_2 = \cdots = \beta_p = 0$ was computed. Power was estimated as the proportion of simulations in which the null hypothesis was rejected at level $\alpha = 0.05$.

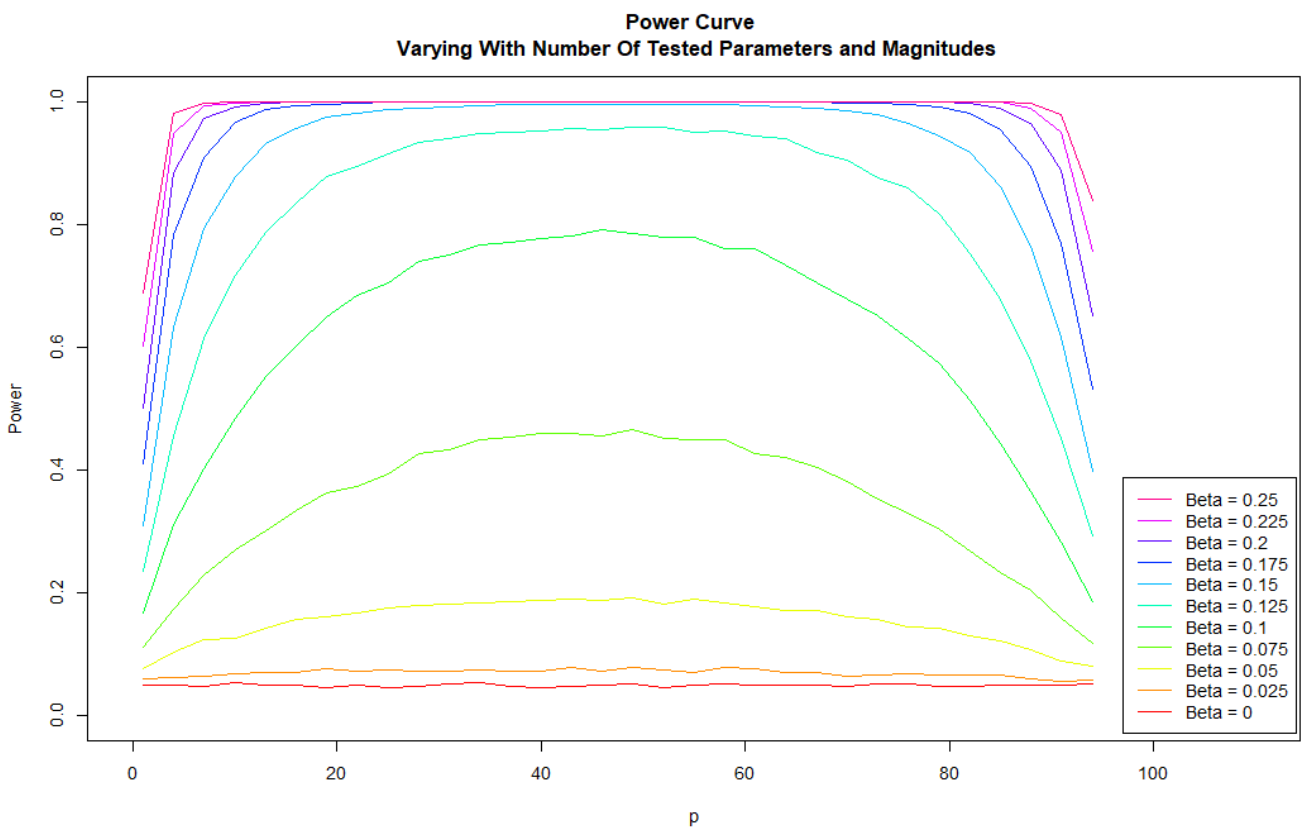

Figure 3: Estimated power of the overall F-test across different numbers of predictors and coefficient magnitudes based on 10,000 simulated datasets.

5.2 Comparing Power When Covariance of Predictors and Parameters are Aligned and Misaligned

To illustrate the phenomenon: how the proposed score test's power depends strongly on how the covariance structure assumed for $\boldsymbol{\beta}$ aligns with the covariance structure of the predictors, we consider a simple simulation with $p = 2$ predictors. This low-dimensional setting allows the geometry of the predictor space and the orientation of the coefficient vector to be easily visualized.

Under Null

Under the null hypothesis, the regression coefficients are set to zero so that the response depends only on the noise term,

$$H_0 : \ \boldsymbol{\beta} = (0, 0)^\mathsf{T}.$$

Data are generated from the standard linear model with independent Gaussian errors.

Under Alternative

Under the alternative hypothesis, the predictors are generated from a correlated Gaussian distribution with covariance matrix

$$\boldsymbol{\Sigma} = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}, \qquad \rho = 0.9,$$

which induces a dominant principal direction of variation approximately along $(1, 1)$. Two different orientations of the coefficient vector are then considered.

In the first configuration, the coefficient vector is aligned with this dominant direction,

$$\boldsymbol{\beta} = (0.10, 0.10)^\mathsf{T},$$

so that the signal lies in the direction where the predictors exhibit the greatest variation.

In the second configuration, the coefficient vector lies approximately orthogonal to this dominant direction,

$$\boldsymbol{\beta} = (0.50, -0.50)^\mathsf{T},$$

placing the signal in a direction where the predictors exhibit relatively little variation.

Data are generated according to

$$\mathbf{Y}_{(n\times 1)} = \mathbf{X}_{(n\times(p+1))}\boldsymbol{\beta}_{((p+1)\times 1)} + \boldsymbol{\epsilon}_{(n\times 1)}, \qquad \boldsymbol{\epsilon} \sim N\left(0, \sigma^2 \boldsymbol{I}\right)$$

where

$$n = 300, \qquad \mathbf{X} \sim N_p(0, \boldsymbol{\Sigma}), \qquad \sigma^2 = 1, \qquad \alpha = 0.05.$$

Two test statistics are compared:

- The classical F-statistic based on the projection matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}\mathbf{X}^\mathsf{T}$.
- A score-type statistic based on $\mathbf{X}\mathbf{X}^\mathsf{T}$ that does not explicitly account for the covariance structure of the predictors.

When the coefficient vector aligns with the dominant direction of variation in $\mathbf{X}$, the score test achieves slightly higher power than the classical F-test. However, when the signal lies in a direction that is poorly represented by the predictor covariance structure, the performance of the score test deteriorates substantially. In contrast, the F-test maintains high power in both configurations because the matrix $(\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}$ explicitly incorporates the geometry of the predictor space. This example illustrates how the performance of the score-based procedure depends critically on the structural assumptions encoded in the prior specification for $\boldsymbol{\beta}$.

5.3 Simulation Demonstrating Invariance

Another important distinction between the classical F-test and the proposed score test concerns invariance. In linear regression with $p < n$, the F-test possesses desirable invariance properties under linear transformations of the predictor variables. In contrast, the score test derived in this framework is not invariant to such transformations because it depends directly on $\mathbf{X}\mathbf{X}^\mathsf{T}$ rather than the projection matrix involving $(\mathbf{X}^\mathsf{T}\mathbf{X})^{-1}$.

To illustrate the practical implications of this property, we conduct a simple simulation with $p = 2$ predictors. Data are generated from the linear model

$$\mathbf{Y}_{(n\times 1)} = \mathbf{X}_{(n\times(p+1))}\boldsymbol{\beta}_{((p+1)\times 1)} + \boldsymbol{\epsilon}_{(n\times 1)}, \qquad \boldsymbol{\epsilon} \sim N\left(0, \sigma^2 \boldsymbol{I}\right)$$

with

$$n = 300, \qquad p = 2, \qquad \sigma^2 = 1$$

The predictors are generated from a Gaussian distribution with covariance matrix

$$\boldsymbol{\Sigma} = \begin{pmatrix} 100 & 0 \\ 0 & 1 \end{pmatrix},$$

so that the predictors are measured on substantially different scales. The coefficient vector is chosen as

$$\boldsymbol{\beta} = 0.02 \cdot (1, 1)^\mathsf{T}$$

Next, a linear transformation is applied to the predictors. Let

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 1/10 \end{pmatrix}$$

We define the transformed design matrix and corresponding coefficients as

$$\mathbf{Z} = \mathbf{X}\mathbf{A}, \qquad \boldsymbol{\gamma} = \mathbf{A}^{-1}\boldsymbol{\beta}$$

This transformation merely rescales the predictors, and therefore the linear predictor remains unchanged,

$$\mathbf{X}\boldsymbol{\beta} = \mathbf{Z}\boldsymbol{\gamma}$$

Consequently, the distribution of $\mathbf{Y}$ and the fitted regression model are identical under the two parameterizations.

For each of 100,000 simulated datasets, both the F-statistic and the score statistic were computed using the original predictors and the transformed predictors. As expected, the F-statistics before and after transformation coincide exactly, reflecting the invariance of the projection matrix $\mathbf{H}$ to nonsingular linear transformations of the predictors. In contrast, the score statistics change substantially under the transformation.